

PFBL: Prototype-Based Fully Balanced Learning for Multimodal Fake News Detection

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Abstract—Multimodal fake news detection (MFND) has received widespread attention. As a typical multimodal task, MFND is troubled by modality imbalance, where the dominant modality suppresses other modalities during the optimization process. Meanwhile, the social credibility maintains the amount of fake news less than that of real news, which causes class imbalance. We call MFND with intertwined influence of modality imbalance and class imbalance as dual imbalanced MFND, i.e., DI-MFND, which has not been well studied. In this article, we propose an approach called prototype-based fully balanced learning (PFBL) for DI-MFND. Specifically, our model contains three main parts. 1) The prototype-based modality-balanced learning (PMB) part, which constructs modality prototypes for each modality. It designs the prototype-based unimodal loss to enhance the intraclass compactness of the poorly performing modality and a constrained gradient optimization strategy to suppress the dominant modality for optimization. 2) The prototype-based class-balanced learning (PCB) part constructs class prototypes. A prototype-oriented discriminative enhancement loss is designed to effectively align samples with corresponding prototypes and increase the inter-class distance, thus enhancing the separability of different categories. 3) The multimodal fusion and classification (MFC) part employs a cross-transformer block to perform interaction between modalities and further integrates features of different modalities by attention mechanism. We propose a joint updating strategy for modality prototypes and class prototypes. Extensive experiments on three widely-used news datasets demonstrate that our approach outperforms state-of-the-art approaches.

Index Terms—Class imbalance, modality imbalance, multimodal fake news detection, prototype learning.

I. INTRODUCTION

WITH the rapid growth of streaming media and big data, online social media platforms such as Twitter and Weibo have gained immense popularity. However, this rapid growth has also led to the widespread dissemination of fake information and news in various forms [18], [28], including text, image, video, and audio. Fake news frequently misleads individuals, thereby influencing their judgments and decisions [15], [38], [56]. Additionally, fake news can be used to distort facts and manipulate public opinion, undermining social trust and stability [3], [10], [23]. To safeguard news accuracy and enhance social media credibility, it is crucial to effectively identify the authenticity of multimodal information shared on these platforms, known as multimodal fake news detection (MFND).

Early researchers combined various modalities with text for MFND [40], [41], [42]. Jin et al. [21] emphasized the utilization of social context for MFND by analyzing user interactions and network structures. Wang et al. [50] introduced a method that combines textual, visual, and social context features to detect fake news. Singhal et al. [43] introduced Spofake to leverage transfer learning to detect fake news by extracting and combining features from various modalities. More recently, Liu et al. [25] proposed an interpretable framework that integrates text, image, and social context features using logic reasoning to detect misinformation. Hu et al. [16] designed a novel unimodal adaptive rationale guidance network that complements small and large LMs by selectively acquiring insights from LLM-generated rationales for small language models. These methods aim to integrate different modal information and adopt appropriate fusion techniques to detect fake news.

However, recent studies have emphasized that the presence of a dominant modality during multimodal training may hinder the optimization of other modalities, ultimately leading to modality imbalance [9], [32], [49]. In this article, we conduct experiments on the Twitter dataset [2] to observe whether there is modality imbalance phenomenon in MFND, with the results presented in Fig. 1(a). Based on the baseline model EANN [50], we also utilize the unimodal versions of it, that is, EANN-Text and EANN-Image, which adopt an independent classifier for each modality to monitor the respective performances of both multimodal and unimodal versions in joint training. Notably, the EANN-Text exhibits similar variations of training accuracy to EANN, whereas

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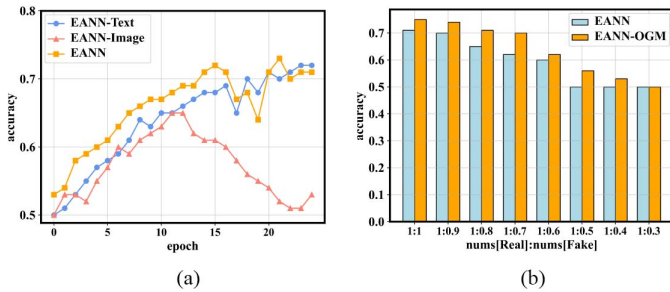


Fig. 1. Influences of modality imbalance and class imbalance issues. (a) Performance of EANN and its uni-modal versions in the training process on Twitter. (b) Performance of EANN and EANN-OGM on Twitter.

the EANN-Image experiences a notable decrease in performance. This suggests that the text modality plays a dominant role during training, suppressing the image modality.

To address modality imbalance issue, Peng et al. [32] proposed on-the-fly gradient modulation (OGM), a method that primarily focuses on dynamically modulating the optimization rates of individual modalities during the training process. Sun et al. [45] directly applied OGM to the field of rumor detection. Chen et al. [5] used ambiguity learning to calculate the ambiguity scores of each modality and adaptively aggregated unimodal features with cross-modal correlation. However, these MFND methods considering modality imbalance primarily focus on utilizing or adjusting a single modality, failing to take into account the joint adjustment of all modalities.

Due to various constraints from society and social media platforms, real news significantly outnumbers fake news. Taking the Twitter dataset as an example, we undersample the fake news samples to observe the performance of model in scenarios with class imbalance. We conduct a comparative experiment between the standard EANN and EANN with OGM. Fig. 1(b) presents the changes in detection accuracy of different models on Twitter. The results reveal that, by accounting for modality imbalance issue, EANN-OGM outperforms the standard EANN in performance. Nevertheless, as the degree of class imbalance increasing, both models exhibit inferior performance. In conclusion, class imbalance significantly undermines the performance of MFND.

To tackle the class imbalance issue in unimodal fake news detection, Bhattacharjee et al. [1] used a bidirectional variational autoencoder to extract informative latent representations from text sequences and then used sampling techniques to resample the latent vectors to address the class imbalance. Hossain et al. [14] employed sample synthesis methods such as SMOTE which increase the number of minority classes to adjust the balance state of the dataset. Keya et al. [22] introduced AugFakeBERT, an approach that uses data augmentation and the BERT model to address the imbalance on fake news datasets. However, these methods focus on unimodal fake news detection task, with a class imbalance in MFND remaining under-explored. Besides, as Fig. 1 illustrates, the intertwined effects of class imbalance and modality imbalance significantly influence the performance of multimodal fake news detection, and considering only one of them is insufficient.

In this article, we propose an approach, called prototype-based fully balanced learning (PFBL) for MFND. Our model contains three parts, namely the prototype-based modality-balanced learning (PMB) part, the prototype-based class-balanced learning (PCB) part, and the Multimodal Fusion and Classification (MFC) part. PMB constructs modality prototypes first, then calculates the ratio of compactness scores for each modality based on the distance between prototypes and corresponding samples. We design one weighted unimodal loss based on the ratio to make the intraclass feature distributions of under-performing modality compact and align the intraclass feature distributions of different modalities. Besides, we design a constrained gradient optimization (CGO) strategy to suppress the dominant modality during the optimization process of multimodal joint training. PCB constructs class prototypes and reduces the distance between class prototypes and corresponding samples while increasing the distance between different class prototypes with the prototype-oriented discriminative enhancement (PODE) loss. MFC employs a cross-transformer module to enable interaction between different modalities and subsequently integrates features of through an attention mechanism. We also designed a joint prototype updating strategy that measures the similarity between modality prototype and class prototype, and the updating process of modality/class prototype is guided by class/modality prototype with high compactness score.

The main contributions of this article are as follows.

- 1) To our knowledge, we are the first work to focus on the intertwined problem of modality imbalance and class imbalance, i.e., DI-MFND. We propose an effective solution called prototype-based fully balanced learning (PFBL) for the intertwined issue.
- 2) We design a weighted prototype-based unimodal loss to make the intraclass feature distribution of under-performing modality compact and introduce a CGO strategy to suppress the dominant modality. We also design the prototype-oriented discriminative enhancement (PODE) loss to reduce the distance between prototypes and corresponding samples, thereby enhancing the separability of different classes. In this way, the intertwined problem of modality imbalance and class imbalance will be effectively alleviated.
- 3) We propose a joint prototype updating strategy for modality prototype and class prototype. We utilize the modality/class prototype with high compactness score to guide the updating of modality/class prototype, and the similarity between different prototypes is used to control the degree of guidance.
- 4) We conduct extensive experiments on three widely used MFND datasets and the results demonstrate that our PFBL outperforms the state-of-the-art methods by 0.032 on Twitter, 0.009 on Fakeddit, and 0.006 on Weibo for accuracy.

II. RELATED WORKS

A. Multimodal Fake News Detection

In recent years, with diversification of the forms of fake news dissemination, there has been a widespread focus on multimodal

fake news detection (MFND) methods [55] that primarily rely on modalities such as images and texts. Wang et al. [50] proposed EANN, which introduces an event discriminator to estimate the differences in feature representations across different events, enabling the model to derive event-invariant features. Singhal et al. [43] introduced Spotfake, which leverages transfer learning to detect fake news by extracting and integrating features across various modalities. Zhang et al. [54] leveraged BERT [7] to extract meaningful features from textual content, capturing both contextual and semantic information, which is useful in domain adaptation. Qian et al. [34] integrated hierarchical semantics of text with multimodal context information, leveraging both textual and visual modalities to enhance detection accuracy. Wei et al. [51] incorporated the concept of event adversarial networks by introducing an event discriminator which allows the model to better generalize to new events, enhancing its ability to detect newly emerging fake news. Peng et al. [31] utilized deep metric learning algorithms to accurately identify fake news by exploiting the deep features of multimodal data. Wang et al. [47] proposed COOLANT which leverages cross-modal contrastive learning to contrast information across different modalities. Focusing on extraction of inconsistent features between modalities, Wu et al. [52] presented HCCIN to comprehensively explore consistent and inconsistent semantics for MFND. Qu et al. [35] integrated image and text features, processed them using quantum convolutional neural network (QCNN) to derive discriminative results. These methods neglect the intertwined issue of modality imbalance and class imbalance.

B. Imbalanced Multimodal Learning

Early research [32], [49] has demonstrated that, in certain cases, the performance of multi-modal models falls short of that of unimodal models. Wang et al. [49] observed that different modalities have different over-fitting rates under a unified optimization strategy, leading to compromised model performances. Peng et al. [32] concluded that modalities with superior performance tend to overshadow the optimization of those with poorer performance, dominating the process of optimization. The above researches reveal that modality imbalance has a significant influence on model performance.

Researchers have proposed several methods to address the issue of modality imbalance [8], [32], [53]. Du et al. [8] utilized the prior knowledge of unimodal models to guide the training of multimodal models, which uses pretrained unimodal models to serve as teachers, providing guiding signals to the multimodal model, ensuring that information from each modality is fully utilized. Peng et al. [32] presented OGM that dynamically adjusts the gradients of different modalities during training by scaling the gradients based on the performance of each modality. Xu et al. [53] proposed a novel loss function designed to balance audio-visual learning by incorporating cosine similarity measures between modalities.

Recently, Chen et al. [5] and Sun et al. [45] take into account the issue of modality imbalance in MFND. Chen et al. [5] quantified the ambiguity by utilizing the distribution divergence among different unimodal features, and adaptively aggregates

unimodal features and cross-modal correlation based on the quantified scores. Sun et al. [45] utilized OGM to alleviate this imbalance by suppressing the optimization of the dominant modality. However, they restrict their focus solely to constraining a single modality with a unified optimization objective, neglecting the influence of other modalities. Furthermore, the intertwined issue of modality imbalance and class imbalance has not been well studied in imbalanced multimodal learning.

C. Prototype Learning

The prototypical network, which was first introduced by [44], has gained significant attention in few-shot learning [11], [19], [26], [37]. Its key concept is that the representations of a class in the feature space are described by a prototype, that is surrounded by or close to multiple samples within that class. In recent years, the concept of prototypes has been widely used to deal with long-tailed recognition [57], multilabel learning [12], class-incremental learning [58], and image classification [6]. This is because prototypes can be used to represent the characteristics of classes, and experiments have proved that such equivalence is effective [9], [24], [44]. In the field of fake news detection, researchers have also proposed several methods utilizing prototypes. Huang et al. [17] incorporated prototypes to capture representative patterns in limited data, leveraging meta-learning and prompts for enhanced detection. Shi et al. [39] employed rough-fuzzy graph learning and prototypes to adaptively model domain shifts, improving fake news detection across diverse contexts. However, they fail to take into account the severe influence of imbalance in MFND. In this article, we aim to address the intertwined issue of modality imbalance and class imbalance for MFND by leveraging the prototype learning technology.

III. OUR APPROACH

A. Model Overview

In this work, we propose an approach called PFBL for MFND. As shown in Fig. 2, PFBL contains three main parts: 1) The prototype-based modality-balanced learning part, which constructs modality prototypes and makes the intraclass feature distribution of under-performing modality compact with the prototype-based unimodal loss. Besides, we design a CGO strategy to suppress the dominant modality in optimization process; 2) The prototype-based class-balanced learning part, which constructs class prototypes, reduces the distance between samples and their respective class prototypes, as well as increasing the distance between different class prototypes by the prototype-oriented discriminative enhancement (PODE) loss; 3) The multimodal fusion and classification part, which employs a cross-transformer block to perform interaction between modalities and further integrates features of different modalities based on attention mechanism. Furthermore, we design a joint prototype updating strategy by taking into account the similarity between modality prototypes and class prototypes.

B. Multimodal Feature Representation

Given the set of news $O = \{o_n\}_{n=1}^N$ includes image and text modalities, where the image modality is represented as

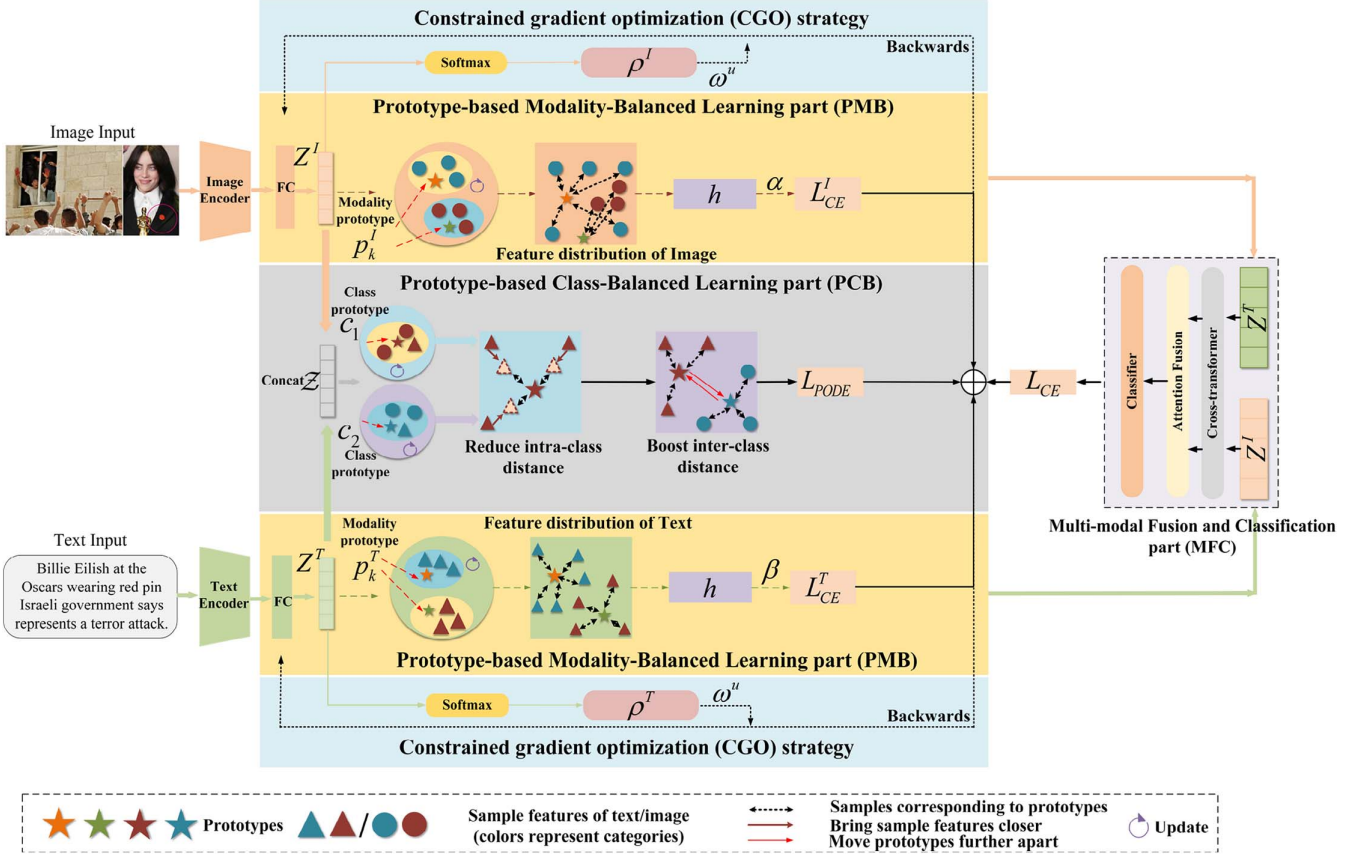


Fig. 2. Architecture of PFB.

$I = \{i_n\}_{n=1}^N$, and the text modality is represented as $T = \{t_n\}_{n=1}^N$. $Y = \{y_n\}_{n=1}^N$ represents the label corresponding to O . BERT [7] is a popular pretrained language model based on Transformer in solving NLP, and we utilize BERT to extract text features. While CLIP [36] is a pretrained model trained on a large dataset, and tremendous experiments have shown its superb robust capability, we utilize CLIP to extract image features. Fully connected layers g^u , where $u \in (T, I)$, are used to map them to the same dimension. $Z^T = \{z_n^T\}_{n=1,2,\dots,N}$ is the text representation set extracted by BERT, $Z^I = \{z_n^I\}_{n=1,2,\dots,N}$ is the image representation set extracted by CLIP.

C. Prototype-Based Modality-Balanced Learning

Modality imbalance is often manifested in the process of multimodal joint training where the dominant modality suppresses the optimization process of other modalities [32], [49], [53], thereby limiting the overall multimodal performance. As shown in Fig. 1(a), the performance of the dominant modality determines the overall performance of the entire model. However, under-performing modality doesn't experience sufficient optimization, resulting in poor performance. Meanwhile, the stretched intraclass feature distribution can result in a decrease in the classification ability of unimodality [46]. In this section, we describe the degree of optimization for each modality from the perspective of feature distribution, and we should enhance the intraclass compactness of the poorly performing modality and

reduce the dispersion in intraclass feature distribution of different modalities.

In this article, we design the prototype-based modality-balanced learning part, which constructs modality prototypes, and measures the degree of compactness of distribution between different modality prototypes and sample features to quantify the performance of each modality. Furthermore, we make the intraclass feature distribution of under-performing modality compact in the distribution with the proposed prototype-based unimodal loss.

1) *Prototype-Based Unimodal Loss:* We denote $Z_k^u = \{z_{k_i}^u\}_{i=1,2,\dots,N_k}$ of Z^u as the subset data of k th category, where N_k is the number of samples in class k , and $k = 1, 2, \dots, M$, M is the number of categories, $u \in (T, I)$. The modality prototype of k th category can be calculated as follows:

$$p_k^u = \frac{1}{N_k} \sum_{i=1}^{N_k} z_{k_i}^u. \quad (1)$$

Then, we leverage modality prototypes to generate intraclass feature distributions of each modality by calculating the distance between the modality prototype and the sample features

$$\mathbf{P}_n^u(y_n = k | z_n^u) = \frac{\exp(-d(z_n^u, p_k^u))}{\sum_{k'} \exp(-d(z_n^u, p_{k'}^u))} \quad (2)$$

where k' is a set of categories that are different from k , and d is a Euclidean distance based distance function. We introduce conditional entropy to further characterize the degree of compactness

of distribution within each modality. The larger the conditional entropy, the greater the stretch of the intraclass feature distribution of the corresponding modality. The conditional entropy based on modality prototype can be calculated as follows:

$$H_n^u = \frac{1}{N} \left[- \sum_{k=1}^M \mathbf{P}_n^u(y_n = k | z_n^u) \log_2 \mathbf{P}_n^u(y_n = k | z_n^u) \right]. \quad (3)$$

H_n^u effectively quantifies the compactness between prototype and samples, and we call H_n^u as compactness score to measure the degree of compactness of distribution between the modality prototype and sample features. In this article, the compactness score denotes the generalized compactness indicator, and the larger value of compactness indicator (compactness score) means less compact in feature distribution. Finally, we calculate the ratio h of compactness scores of different modalities

$$h = \frac{\sum_{i \in B} H_i^T}{\sum_{i \in B} H_i^I} \quad (4)$$

where B is a random mini-batch. We use h to measure the discrepancy of compactness of distribution between two modalities. For a large h , the compactness score of text modality will be large, indicating the intraclass feature distribution of the text modality will be less compact, and vice versa. Therefore, we introduce one prototype-based unimodal loss for each modality, to enhance the compactness of distribution for the under-performing modality. The loss can be formulated as

$$L_{CE}^u = \mathbb{E}_{\mathbf{P}(z_n^u, y_n)} \left[- \sum_{n=1}^N \log \frac{\exp(-d(z_n^u, p_{y_n}^u))}{\sum_{k=1}^M \exp(-d(z_n^u, p_k^u))} \right]. \quad (5)$$

We use L_{PUL} to represent the total prototype-based unimodal loss for each modality as follows:

$$L_{PUL} = \alpha L_{CE}^T + \beta L_{CE}^I \quad (6)$$

where α and β are parameters used to control the magnitude of enhancement, and we utilize h to adjust α and β accordingly

$$\begin{cases} \alpha = 0, \beta = \frac{1}{h} - 1, & h < 1 \\ \alpha = 1 - \frac{1}{h}, \beta = 0, & \text{otherwise.} \end{cases} \quad (7)$$

A large h means the intraclass feature distribution of the text modality will be less compact. In this case, we should consider the regulation of the text modality. As shown in (7), we define an adjustment value $1 - (1/h)$ to adjust text modality, and we assign 0 to β , which controls the adjustment of image modality. Contrastively, a small h means the intraclass feature distribution of the text modality will be more compact. In this case, we should not consider the regulation of the text modality. As shown in (7), we assign 0 to α , which controls the adjustment of text modality. And we define an adjustment value $(1/h) - 1$ to adjust image modality. With the prototype-based unimodal loss, we ensure the compactness of intraclass feature distribution of poor-performing modality.

2) *CGO Strategy*: Even though we make the intraclass feature distribution of under-performing modality compact, it does

not prevent the dominant modality from disrupting the optimization process. Hence, measures must be taken to restrain the dominant modality. In this part, we design a CGO strategy for model's gradient update in a certain magnitude determined by contributions.

Specifically, we first calculate the contribution of each modality toward the optimization objective, which can be formulated as

$$s_n^u = \sum_{m=1}^M 1_{m=y_n} \text{softmax} \left(W^u z_n^u + \frac{b}{2} \right)_m \quad (8)$$

where W^u and $b/2$ are the weight and bias of classification layers which will be introduced in Section III-E. After calculating the contribution of each modality, we calculate the ratio ρ^u of modal contributions of different modalities as follows:

$$\rho^u = \frac{\sum_{i \in B} s_i^u}{(\sum_{q \in \{T, I\}} \sum_{i \in B} s_i^q) / 2}. \quad (9)$$

Then, we calculate the balance factor ω^u as follows:

$$\omega^u = \begin{cases} 1 - \tanh(\rho^u), & \text{if } \rho^u > 1 \text{ and } \tilde{\rho} \in (0.2, 0.8) \\ 1, & \text{otherwise} \end{cases} \quad (10)$$

where $\tilde{\rho}$ is a threshold to regulate the magnitude of optimization. $\tilde{\rho}$ is defined as follows:

$$\tilde{\rho} = \frac{|\sum_{i \in B} s_i^T - \sum_{i \in B} s_i^I|}{(\sum_{q \in \{T, I\}} \sum_{i \in B} s_i^q) / 2} \quad (11)$$

where $|\cdot|$ is the absolute value sign. $|\sum_{i \in B} s_i^T - \sum_{i \in B} s_i^I|$ is calculated to measure the difference of modal contributions of different modalities. If the difference is insufficiently small, adjusting the optimization process is unnecessary; conversely, if the magnitude is excessively large, excessive suppression can alter the optimization direction. In this article, we experimentally set a threshold range from 0.2 to 0.8.

Finally, we integrate ω^u into the SGD optimization algorithm, the optimization process is as follows:

$$\theta_{t+1}^u = \theta_t^u - \omega^u \tilde{g}^u(\theta_t^u) \quad (12)$$

where θ^u is the parameter of g^u , and t represents the t th iteration.

By using CGO, we inhibit the optimization of the modalities with better performance, while the modalities with poor performance are not affected.

By leveraging prototype-based unimodal loss to make the intraclass feature distribution of under-performing modality compact, while suppressing the dominant modality through a gradient modulation strategy, our PMB simultaneously imposes balance constraints on all modalities.

D. Prototype-Based Class-Balanced Learning

As discussed above in Section I, both modality imbalance and class imbalance have significant influences on MFND. Previous studies [13], [48] have shown that one of the main reasons for class imbalance is the stretched feature distribution in a unified space. As the degree of compactness between the prototype and sample features mirrors the compactness of the feature

distribution, we utilize prototype to tackle the challenge of class imbalance.

For this part, we propose the prototype-based class-balanced learning part specifically addressing class imbalance. We first construct class prototypes for each category. For different class prototypes, we enlarge the distance between them to enhance the discriminative ability of the model. For each class prototype, we minimize the distance between the class prototype and its corresponding sample features while maximizing the distance between prototype and sample features of other categories, which ensures the compactness of intraclass feature distribution of each class prototype.

Specifically, we map $z_{k_i}^T$ and $z_{k_i}^I$ into a unified space and concatenate them as $\mathcal{Z}_{k_i} = \{\tilde{z}_{k_i}\}_{i=1,2,\dots,N_k}$. Then we map $\tilde{z}_{k_i}$ to the same dimension as $z_{k_i}^u$, which can be represented as z_{k_i} . The class prototype of the k th category can be calculated as follows:

$$c_k = \frac{1}{N_k} \sum_{i=1}^{N_k} z_{k_i}. \quad (13)$$

Then, we introduce a prototype-oriented discriminative enhancement (PODE) loss, which brings the class prototype closer to its corresponding sample features, while pushing the sample features of other class prototypes away. Meanwhile, the prototypes of different categories are also pushed apart. The loss L_{PODE} is calculated as follows:

$$L_{\text{PODE}} = \sum_{k=1}^M \sum_{i=1}^{N_k} \log \left[\frac{\exp(-d(z_{k_i}, c_k))}{\sum_{j \in \mathbb{k}} \exp(-d(z_{k_i}, c_j))} \right] - \frac{1}{M(M-1)} \sum_{k=1}^M \sum_{j \in \mathbb{k}} \|c_k - c_j\|_2^2 \quad (14)$$

where $\mathbb{k} = \{j | 1 \leq j \leq M, j \neq k\}$. L_{PODE} aims to align the class samples with their respective class prototypes, simultaneously, and different class prototypes will be pushed further apart. With L_{PODE} , we can not only compactify the feature space distribution within each category, but also enhance the model's discriminative ability, thereby effectively alleviating the issue of class imbalance.

E. Multimodal Fusion and Classification

To obtain better multimodal fused features and eliminate differences of each modality, we design the multimodal fusion and classification part for fusing and classifying multimodal fake news data.

Specifically, for Z^T and Z^I , we first utilize cross-modal transformer [29] to achieve interaction between modalities, which is formulated as

$$\tilde{Z}^T, \tilde{Z}^I = \text{cross-transformer}(Z^T, Z^I, \theta_{\text{CT}}) \quad (15)$$

where θ_{CT} is the parameter of the cross-modal transformer. Then, we employ an attention mechanism to fuse the features and obtain the final fused features, the attention coefficients can be calculated as

$$\begin{aligned} \delta^T &= f(\tilde{Z}^T, \theta_T) \\ \delta^I &= f(\tilde{Z}^I, \theta_I) \end{aligned} \quad (16)$$

where $f(\cdot)$ represents the attention network, δ^T and δ^I are the attention coefficients of $\tilde{Z}^T$ and $\tilde{Z}^I$, θ_T and θ_I are the parameters of $f(\cdot)$. Finally, we splice them to obtain final feature representation Z_{fuse}

$$Z_{\text{fuse}} = [\delta^T \tilde{Z}^T; \delta^I \tilde{Z}^I]. \quad (17)$$

Then, we feed Z_{fuse} into a classifier composed of a one-layer MLP utilizing a ReLU activation function, and obtain cross-entropy loss as classification loss, which can be represented as L_{CE} . The total loss will be

$$L = L_{\text{CE}} + \lambda_1 L_{\text{PUL}} + \lambda_2 L_{\text{PODE}}. \quad (18)$$

F. Joint Prototype Updating Strategy

Traditional prototype update strategies independently update different kinds of prototypes. If we adopt this manner, it lacks the ability to jointly consider the impact of modality imbalance and class imbalance. It might only focus on one aspect, such as updating modality prototypes without considering how it impacts class-related features, or vice versa. This one-sided approach fails to address the complex and intertwined nature of the two imbalance problems in MFND. In this section, we introduce a joint updating strategy for modality prototype and class prototype, aiming to obtain prototypes that are more conducive to solving the intertwined issue of modality imbalance and class imbalance. For modality prototype, we employ a common and effective updating strategy first. The detailed updating process of p_k^u for $(t+1)$ can be formulated as follows:

$$(p_k^u)_{t+1} = (1 - \varepsilon)(p_k^u)_t + \varepsilon(p_k^u)_{t+1} \quad (19)$$

where ε is a hyperparameter.

To further explore the intricate relationship between modality prototypes and class prototypes, we enhance (19) by leveraging the similarity relationships between them. Specifically, we ensure that the modality prototype is similar to the class prototype that has a compact intraclass feature distribution.

First, we need to find out the class prototype which corresponds to the compact intraclass feature distribution. We emulate (2) to compute the intraclass feature distribution based on class prototype, which can be formulated as follows:

$$\mathbf{P}_i(y_i = k | z_{k_i}) = \frac{\exp(-d(z_{k_i}, c_k))}{\sum_{k'} \exp(-d(z_{k_i}, c_{k'}))} \quad (20)$$

where k' is a set of categories that are different from k . Then, we emulate (4) to calculate the ratio h^k to observe the intraclass feature distribution of k th category

$$h^k = \frac{\sum_{i \in B} \mathbf{P}_i(y_i = k | z_{k_i})}{\sum_{i \in B} \mathbf{P}_i(y_i = k' | z_{k_i})}. \quad (21)$$

After selecting the suitable class prototype that corresponds to the compact intraclass feature distribution, we calculate the similarity between modality prototype and the chosen class prototype in (22). The cosine similarity is used

for similarity calculation

$$\text{sim}(p_k^u, c_k) = \frac{p_k^u \cdot c_k}{|p_k^u| |c_k|} \quad (22)$$

$$\begin{cases} \eta_1 = 1 + \frac{1}{\text{sim}(p_k^u, c_k)}, & \text{if } h^k > 1 \\ \eta_2 = 1 + \frac{1}{\text{sim}(p_k^u, c_{k'})}, & \text{otherwise.} \end{cases} \quad (23)$$

In (23), we calculate the weights for each modality prototype. Finally, we improve (19) by dynamically updating the weights of modality prototype based on the similarity relationships between modality prototype and the chosen class prototype with high compactness. The final updating process of modality prototype is as follows:

$$\begin{cases} (p_k^u)_{t+1} = (1 - \varepsilon)(p_k^u)_t + \varepsilon\eta_1(c_k)_{t+1}, & \text{if } h^k > 1 \\ (p_k^u)_{t+1} = (1 - \varepsilon)(p_k^u)_t + \varepsilon\eta_2(c_{k'})_{t+1}, & \text{otherwise} \end{cases} \quad (24)$$

where for the modality prototype exhibiting high similarity to the suitable class prototype, we assign a relatively small weight to the class prototype. For the modality prototype with lower similarity, we assign a large weight to the class prototype.

For class prototype, we integrate the modality prototype into the updating process of class prototype. Considering h in (4) indicates ratio of compactness score of two modalities, we enable the modality prototype that corresponds to the compact intraclass feature distribution to guide the updating of class prototype. The process of the update can be described as

$$\begin{cases} (c_k)_{t+1} = (1 - \delta)(c_k)_t + \delta(p_k^T)_{t+1}, & \text{if } h < 1 \\ (c_k)_{t+1} = (1 - \delta)(c_k)_t + \delta(p_k^I)_{t+1}, & \text{otherwise} \end{cases} \quad (25)$$

where δ is a hyperparameter.

By adopting the joint prototype updating strategy, the interaction between modality prototypes and class prototypes is achieved, resulting in compact intraclass feature distributions, which further effectively alleviates the intertwined issues of modality imbalance and class imbalance.

The overall model optimization process is summarized as Algorithm 1.

IV. EXPERIMENTS

A. Datasets

We validate our model on three widely used datasets, Weibo [20], Twitter [2], and Fakeddit [30].

Weibo Dataset: Weibo is specifically tailored for Chinese-language fake news detection, and it contains a significant number of posts from the popular Chinese microblogging platform. It typically includes features such as post content, timestamps, user information, and engagement metrics. We divide this dataset into a training set and a testing set with a ratio of 8:2, where the training set contains 7532 news items and the testing set contains 1996 news items.

Twitter Dataset: Twitter focuses on English-language fake news detection. It comprises a vast collection of tweets, user

Algorithm 1: Prototype-Based Fully Balanced Learning.

Input data $O = \{o_n\}_{n=1}^N$, corresponding labels $Y = \{y_n\}_{n=1}^N$, initialized network parameters, epoch number E ;
 $e = 0$;
while $e < E$ **do**
 Calculate modality prototypes p_k^u and class prototypes c_k by using Eqs. (1) and (13);
 for mini-batch data B at step t **do**
 Obtain the representations $z_{k_i}^u$ and z_{k_i} ;
 Calculate H_n^u and h using Eqs. (3) and (4);
 Calculate ρ^u and $\tilde{\rho}$ using Eqs. (9) and (10);
 Calculate L_{PUL} using Eq. (5), Eq. (6) and Eq. (7);
 Calculate L_{PODE} using Eq. (14);
 Calculate the final loss L and update the model using Eq. (18);
 Update θ_{t+1}^u using Eq. (10) and Eq. (12);
 Update $(p_k^u)_{t+1}$ and $(c_k)_{t+1}$ using Eqs. (24) and Eq. (25);
 $e = e + 1$;

TABLE I
STATISTICS OF THREE DATASETS

Statistics		Weibo	Twitter	Fakeddit
Training Set	Fake	3749	6125	15 000
	Real	3783	4477	15 000
Testing Set	Fake	1000	1531	5000
	Real	996	1120	5000
All		9528	13 253	40 000

details, tweet metadata, and original images or videos attached to the tweets. We divide the dataset into a training set and a testing set, where the training set contains 8617 news items and the testing set contains 2059 news items.

Fakeddit Dataset: Fakeddit is a huge dataset which is specifically designed for analyzing and understanding the characteristics of misleading information on Reddit. Following [27], we randomly select 30 000 image-text pairs for training and 10 000 image-text pairs for testing.

The detailed information of three datasets is listed in Table I.

B. Baselines

We compare the following baselines with our model.

- 1) *EANN* [50], which is a GAN-based event adversarial neural network designed for MFND.
- 2) *SpotFake* [43], which utilizes BERT and VGG-19 to extract features from text and images and concatenates them for classification.
- 3) *BDANN* [54], which proposes a BERT-based domain adaptation network by fusing multimodal data.
- 4) *HMCAN* [34], which integrates hierarchical multimodal contextual attention to jointly model text and image representations.

TABLE II
PERFORMANCE COMPARISON BETWEEN OUR MODEL AND BASELINE METHODS ON WEIBO, TWITTER, AND FAKEDDIT DATASETS

Dataset	Method	Accuracy	Fake News			Real News		
			Precision	Recall	F1-score	Precision	Recall	F1-score
Weibo	EANN [KDD-2018]	0.782 ± 0.004	0.827 ± 0.004	0.697 ± 0.006	0.756 ± 0.005	0.753 ± 0.004	0.863 ± 0.003	0.804 ± 0.003
	SpotFake [BigMM-2019]	0.870 ± 0.003	0.855 ± 0.005	0.892 ± 0.004	0.873 ± 0.003	0.769 ± 0.003	0.807 ± 0.004	0.787 ± 0.003
	BDANN [IJCNN-2020]	0.842 ± 0.004	0.830 ± 0.003	0.870 ± 0.004	0.850 ± 0.003	0.850 ± 0.003	0.820 ± 0.007	0.830 ± 0.005
	HMCAN [SIGIR-2021]	0.885 ± 0.004	0.920 ± 0.004	0.845 ± 0.003	0.881 ± 0.006	0.856 ± 0.006	0.926 ± 0.003	0.890 ± 0.006
	MEAN [SPL-2022]	0.894 ± 0.004	0.900 ± 0.003	0.870 ± 0.004	0.890 ± 0.004	0.890 ± 0.004	0.910 ± 0.004	0.900 ± 0.003
	CAFE [WWW-2022]	0.898 ± 0.005	0.922 ± 0.006	0.881 ± 0.004	0.901 ± 0.005	0.874 ± 0.017	0.917 ± 0.005	0.895 ± 0.004
	MRML [ICASSP-2023]	0.897 ± 0.008	0.896 ± 0.006	0.905 ± 0.004	0.901 ± 0.005	0.898 ± 0.004	0.887 ± 0.006	0.892 ± 0.005
	COOLANT [ACM MM-2023]	0.923 ± 0.006	0.927 ± 0.005	0.923 ± 0.008	0.925 ± 0.006	0.919 ± 0.008	0.922 ± 0.005	0.920 ± 0.006
	QMFND [INFORM FUSION-2024]	0.869 ± 0.004	0.900 ± 0.004	0.810 ± 0.004	0.850 ± 0.004	0.840 ± 0.006	0.920 ± 0.004	0.880 ± 0.005
	PFBL	0.929 ± 0.003	0.940 ± 0.002	0.930 ± 0.003	0.935 ± 0.004	0.922 ± 0.003	0.938 ± 0.003	0.930 ± 0.004
Twitter	EANN [KDD-2018]	0.648 ± 0.004	0.810 ± 0.005	0.498 ± 0.007	0.617 ± 0.005	0.584 ± 0.004	0.759 ± 0.004	0.660 ± 0.006
	SpotFake [BigMM-2019]	0.790 ± 0.004	0.786 ± 0.004	0.747 ± 0.004	0.766 ± 0.004	0.793 ± 0.007	0.827 ± 0.006	0.810 ± 0.004
	BDANN [IJCNN-2020]	0.830 ± 0.005	0.810 ± 0.004	0.630 ± 0.006	0.710 ± 0.006	0.830 ± 0.002	0.930 ± 0.003	0.880 ± 0.005
	HMCAN [SIGIR-2021]	0.897 ± 0.004	0.971 ± 0.006	0.801 ± 0.003	0.878 ± 0.003	0.853 ± 0.002	0.979 ± 0.005	0.912 ± 0.005
	MEAN [SPL-2022]	0.780 ± 0.003	0.690 ± 0.004	0.840 ± 0.005	0.760 ± 0.004	0.870 ± 0.003	0.740 ± 0.003	0.800 ± 0.003
	CAFE [WWW-2022]	0.917 ± 0.003	0.964 ± 0.006	0.878 ± 0.006	0.919 ± 0.004	0.872 ± 0.004	0.963 ± 0.004	0.915 ± 0.003
	MRML [ICASSP-2023]	0.803 ± 0.005	0.777 ± 0.005	0.747 ± 0.006	0.762 ± 0.006	0.821 ± 0.005	0.844 ± 0.005	0.832 ± 0.006
	COOLANT [ACM MM-2023]	0.900 ± 0.005	0.879 ± 0.005	0.922 ± 0.004	0.900 ± 0.005	0.923 ± 0.004	0.880 ± 0.004	0.901 ± 0.004
	QMFND [INFORM FUSION-2024]	0.918 ± 0.004	0.880 ± 0.003	0.970 ± 0.004	0.920 ± 0.003	0.970 ± 0.003	0.870 ± 0.003	0.910 ± 0.004
	PFBL	0.950 ± 0.003	0.930 ± 0.003	0.980 ± 0.003	0.950 ± 0.003	0.980 ± 0.002	0.920 ± 0.003	0.950 ± 0.003
Fakeddit	EANN [KDD-2018]	0.724 ± 0.003	0.727 ± 0.006	0.719 ± 0.003	0.723 ± 0.004	0.722 ± 0.004	0.729 ± 0.006	0.726 ± 0.004
	SpotFake [BigMM-2019]	0.819 ± 0.004	0.801 ± 0.004	0.848 ± 0.005	0.824 ± 0.005	0.839 ± 0.003	0.790 ± 0.004	0.813 ± 0.004
	BDANN [IJCNN-2020]	0.812 ± 0.004	0.836 ± 0.006	0.776 ± 0.003	0.805 ± 0.004	0.791 ± 0.003	0.847 ± 0.004	0.818 ± 0.003
	HMCAN [SIGIR-2021]	0.881 ± 0.003	0.880 ± 0.006	0.882 ± 0.004	0.881 ± 0.003	0.882 ± 0.006	0.880 ± 0.003	0.881 ± 0.004
	MEAN [SPL-2022]	0.910 ± 0.003	0.930 ± 0.004	0.890 ± 0.004	0.910 ± 0.005	0.890 ± 0.004	0.930 ± 0.003	0.910 ± 0.003
	CAFE [WWW-2022]	0.912 ± 0.005	0.946 ± 0.003	0.886 ± 0.006	0.916 ± 0.005	0.878 ± 0.004	0.942 ± 0.004	0.909 ± 0.004
	MRML [ICASSP-2023]	0.840 ± 0.006	0.819 ± 0.005	0.874 ± 0.005	0.846 ± 0.004	0.865 ± 0.005	0.807 ± 0.005	0.835 ± 0.004
	COOLANT [ACM MM-2023]	0.925 ± 0.003	0.930 ± 0.004	0.920 ± 0.003	0.925 ± 0.003	0.910 ± 0.004	0.880 ± 0.003	0.915 ± 0.003
	QMFND [INFORM FUSION-2024]	0.942 ± 0.004	0.930 ± 0.005	0.950 ± 0.005	0.940 ± 0.004	0.950 ± 0.005	0.930 ± 0.004	0.940 ± 0.004
	PFBL	0.951 ± 0.003	0.955 ± 0.004	0.945 ± 0.003	0.950 ± 0.004	0.945 ± 0.004	0.955 ± 0.005	0.950 ± 0.004

Note: The bold entries denote the best results among multiple compared methods.

- 5) *MEAN* [51], which designs a multimodal generator and a dual discriminator for fake news detection.
- 6) *CAFE* [5], which quantifies the ambiguity between text and images by estimating the dispersion of feature distributions and adaptively integrates unimodal features with cross-modal correlation.
- 7) *MRML* [31], which analyzes news content by constructing a multidimensional relationship graph, and leverages graph embedding techniques to get low-dimensional representations.
- 8) *COOLANT* [47], which proposes a cross-modal contrastive learning framework to explore the correlation and consistency between modalities and analyze information between two modalities from a complementary perspective.
- 9) *QMFND* [35], which integrates multimodal features and passes them through a quantum convolutional neural network (QCNN) to obtain discriminative results.

C. Experimental Settings

We introduce the important experimental settings as follows.

Evaluation Metrics: We use accuracy as our evaluation metric, which is the most commonly used indicator for classification task. Additionally, to take class imbalance into account, we also employ evaluation metrics such as precision, recall, and F-measure (F1).

Implementation Details: We employ the pretrained models BERT and CLIP without fine-tuning to extract text and image features, respectively. The batch-size for all three datasets is set to 64.

The maximum number of epochs is 100, and we adopt the early stopping mechanism at 85th epoch. The learning rate is 0.001. The optimizer is the SGD with 0.001 weight decay. We employ a grid search method to fine-tune the hyperparameters in (18), (24), and (25). The search range for λ_1 , λ_2 , ε , and δ is [0, 1]. Ultimately, we set λ_1 to 0.4, λ_2 to 0.6, ε to 0.8 and δ to 0.2 for all three datasets.

D. Experimental Results

Table II shows the performance with standard deviation of our model and baseline methods on the Weibo, Twitter, and Fakeddit datasets. The results of the baseline methods are reported by running their models on the dataset in this article. As shown in Table II, our PFBL significantly outperforms all compared methods in terms of both accuracy and F1 on three datasets, which demonstrates the effectiveness of our model. Specifically, PFBL achieves remarkable results with the accuracy of 0.929, 0.950, and 0.951 on Weibo, Twitter, and Fakeddit datasets. In terms of F1, PFBL gains a 0.010 (0.935–0.925) improvement on fake news and 0.010 (0.930–0.920) improvement on real news respectively on Weibo, 0.030 (0.950–0.920) improvement on fake news and 0.035 (0.950–0.915) improvement on real news respectively on Twitter, 0.010 (0.950–0.940) improvement on fake news and 0.010 (0.950–0.940) improvement on real news respectively on Fakeddit.

The reason for PFBL's improvement is as follows: For EANN and SpotFake, these methods primarily focus on event-robust feature learning and feature concatenation, respectively.

TABLE III
P-VALUES OF PFBL AND OTHER COMPARED METHODS ON THREE DATASETS

Dataset	PFBL								
	EANN	SpotFake	BDANN	HMCAN	MEAN	CAFE	MRML	COOLANT	QMFND
Weibo	$1.117 \times e-5$	$6.000 \times e-6$	$1.412 \times e-8$	$2.233 \times e-3$	$7.122 \times e-7$	$1.812 \times e-6$	$4.119 \times e-8$	$4.576 \times e-6$	$5.000 \times e-5$
Twitter	$1.422 \times e-5$	$1.732 \times e-9$	$5.221 \times e-6$	$4.321 \times e-9$	$8.224 \times e-10$	$1.227 \times e-12$	$6.350 \times e-7$	$1.231 \times e-5$	$1.329 \times e-4$
Fakeddit	$1.200 \times e-7$	$2.134 \times e-5$	$1.763 \times e-6$	$1.172 \times e-9$	$7.325 \times e-8$	$7.000 \times e-6$	$4.159 \times e-9$	$4.000 \times e-12$	$1.200 \times e-8$

They lack the sophisticated mechanisms to handle imbalance learning. For BDANN and HMCAN, these methods incorporate domain adaptation and hierarchical attention, but they do not specifically target the dual imbalance issue. For COOLANT and QMFND, although they perform cross-modal contrastive learning and quantum convolutional networks, they do not integrate a comprehensive strategy to handle both modality and class imbalance problems. CAFE [5] takes into account modality imbalance and uses ambiguity learning to calculate the ambiguity scores of each modality and adaptively aggregates unimodal features with cross-modal correlation. However, none of the baseline methods including CAFE consider the intertwined issue of modality imbalance and class imbalance, overlooking their severe influence on MFND. For modality imbalance, PFBL employs prototype-based unimodal loss to enhance the intraclass compactness of the poorly performing modality while suppressing dominant modality during optimization by CGO strategy. For class imbalance, our PFBL utilizes a prototype-oriented discriminative enhancement loss to maintain a compact intraclass feature distribution and enhance discriminability between different classes. Moreover, PFBL's joint prototype updating strategy achieves the interplay between modality prototypes and class prototypes, resulting in prototypes with stronger discriminability. PFBL effectively tackles the intertwined problem and achieves the best performance.

To statistically analyze the difference between our approach and compared methods, we conduct the Kruskal Wallis test [4] at significance level of 0.05 to perform statistical significance test between PFBL and other methods. Table III shows the P-value at significance level of 0.05 between PFBL and other methods. When P-value is lower than 0.05, we can consider that PFBL has a significant difference against the corresponding comparison methods. From Table III, we can find the P-values are significantly lower than 0.05. This demonstrates PFBL indeed has a significant difference from the comparison methods.

In addition, we have performed T-SNE to observe the feature distributions on Weibo, Twitter, and Fakeddit. Fig. 3 shows the T-SNE diagrams of sample feature distribution on Weibo, Twitter, and Fakeddit, respectively. For original sample distribution, we simply concatenate the original features of the text and image. In Fig. 3(a), 3(c), and 3(e), we can see that real and fake news mix together, whereas in Fig. 3(b), 3(d), and 3(f), we can find that real and fake news exhibit relatively good separability, with few samples are not completely separated. This demonstrates the effectiveness of PFBL for multimodal fake news classification.

E. Ablation Studies

To evaluate the effectiveness of each part in our proposed PFBL, we individually remove each part for comparison. We

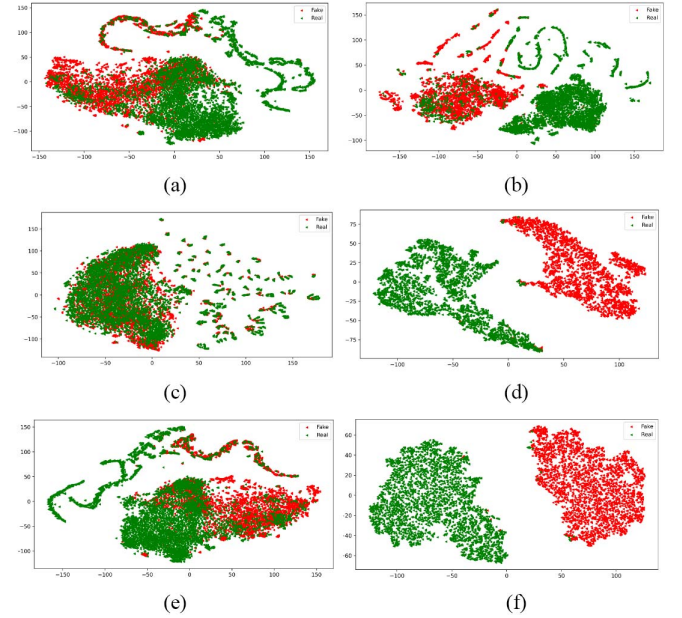


Fig. 3. Feature distribution on three datasets. (a) Distribution of original samples on Weibo. (b) Distribution of learned features on Weibo. (c) Distribution of original samples on Twitter. (d) Distribution of learned features on Twitter. (e) Distribution of original samples on Fakeddit. (f) Distribution of learned features on Fakeddit.

TABLE IV
ABLATION STUDIES OF PMB, PCB, AND MFC ON THREE DATASETS

Dataset	Method	Accuracy	F1-Score	
			Fake News	Real News
Weibo	PFBL	0.929	0.935	0.930
	-w/o PMB	0.910	0.915	0.915
	-w/o PCB	0.913	0.920	0.910
	-w/o MFC	0.920	0.920	0.920
Twitter	PFBL	0.950	0.950	0.950
	-w/o PMB	0.930	0.940	0.920
	-w/o PCB	0.926	0.920	0.930
	-w/o MFC	0.934	0.930	0.940
Fakeddit	PFBL	0.951	0.950	0.950
	-w/o PMB	0.935	0.940	0.930
	-w/o PCB	0.938	0.940	0.940
	-w/o MFC	0.937	0.940	0.940

Note: The bold entries denote the best results among multiple compared methods.

removed the PMB part, which we call it as *PFBL w/o PMB*. We remove the PCB part, which is named *PFBL w/o PCB*. We removed the MFC part, and call it *PFBL w/o MFC*. For *PFBL w/o PMB* and *PFBL w/o PCB*, we use (19) to update both modality prototype and class prototype. For *PFBL w/o MFC*, we simply concatenate Z^T and Z^I for classification. Table IV shows the results of ablation studies of PMB, PCB, and MFC.

TABLE V
ABLATION STUDIES OF OTHER VARIANTS OF PFBL ON THREE DATASETS

Dataset	Method	Accuracy	F1-Score	
			Fake News	Real News
Weibo	PFBL	0.929	0.935	0.930
	-w/o PUL	0.915	0.910	0.920
	-w/o CGO	0.920	0.930	0.920
	-w/o JPU	0.910	0.910	0.910
	-w/o PB	0.818	0.790	0.840
Twitter	PFBL	0.950	0.950	0.950
	-w/o PUL	0.930	0.930	0.930
	-w/o CGO	0.936	0.930	0.940
	-w/o JPU	0.920	0.930	0.920
	-w/o PB	0.802	0.800	0.800
Fakeddit	PFBL	0.951	0.950	0.950
	-w/o PUL	0.935	0.920	0.940
	-w/o CGO	0.937	0.940	0.940
	-w/o JPU	0.930	0.930	0.930
	-w/o PB	0.890	0.910	0.880

Note: The bold entries denote the best results among multiple compared methods.

From the table, the performance of all variants is obviously worse than the original PFBL on three datasets. These results mean that the designed prototype-based modality-balanced learning part, the prototype-based class-balanced learning part, and the multimodal fusion and classification part are useful for multimodal fake news detection. Furthermore, for the results of *PFBL w/o PMB* and *PFBL w/o PCB*, merely considering one imbalance issue without addressing the intertwined problem of modality imbalance and class imbalance is insufficient. This also demonstrates the effectiveness of PFBL in jointly addressing this intertwined issue.

To further verify the effect of the prototype-based unimodal loss and CGO strategy in the PMB part of the model, we separately remove them, which we call *PFBL w/o PUL* and *PFBL w/o CGO*. To demonstrate the effectiveness of the joint prototype updating strategy, we remove it, which we call *PFBL w/o JPU*, and we use (19) to update both modality prototype and class prototype. To validate the effectiveness of the combination of the PMB and the PCB, we remove them, which we call *PFBL w/o PB*. Table V shows the results of ablation studies of variants above.

From the table, the performance of all variants is worse than the original PFBL. For *PFBL w/o PUL* and *PFBL w/o CGO*, the prototype-based unimodal loss and CGO strategy are both effective separately in terms of strengthening the compactness of intra-class feature distribution for under-performing modality and suppressing dominant modality during the optimization process. For *PFBL w/o JPU*, the joint prototype updating strategy is superior to the traditional prototype update strategy. For *PFBL w/o PB*, it demonstrates the necessity of jointly considering the intertwined issue of modality imbalance and class imbalance.

F. Discussion

1) *Effects of Modality-Balanced Learning*: To further demonstrate the effectiveness of our PFBL in addressing modality imbalance, considering ρ^u represents the differences of contributions of different modalities, we observe the variations of ρ^u in (9). Fig. 4

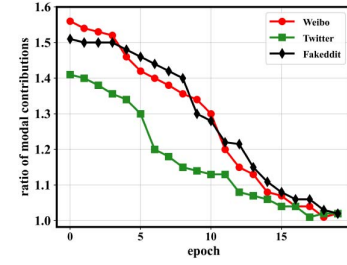


Fig. 4. Trend curve of ρ^u on three datasets.

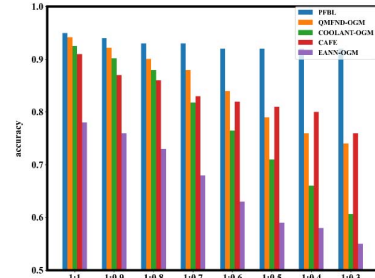


Fig. 5. Analysis of class imbalance on Fakeddit.

shows the trend curve of ρ^u on three datasets. It can be observed that as the training process progresses, ρ^u gradually decreases, ultimately achieving balance. The results demonstrate that our PMB is effective in achieving modality balance state.

2) *Effects of Class-Balanced Learning*: To observe whether the class imbalance problem has been effectively alleviated, we conduct a class imbalance analysis for PFBL, along with QMFND with OGM (QMFND-OGM), COOLANT with OGM (COOLANT-OGM), CAFE, and EANN with OGM (EANN-OGM) on the Fakeddit dataset. Notably, QMFND-OGM, COOLANT-OGM, CAFE, and EANN-OGM account for modality imbalance issue but neglecting class imbalance. To observe the performance of the model under different class imbalance states, we maintain the number of real news samples in the Fakeddit dataset and undersample the fake news samples to obtain different class imbalance states. Fig. 5 shows the results. It can be observed that as the degree of class imbalance increases, five methods exhibit a downward trend. However, even in extreme cases of imbalance such as 1:0.3, PFBL maintains a relatively high accuracy compared with other methods, which further demonstrates the effectiveness of PCB in dealing with class imbalance issue.

3) *Visualization of Prototype Evolvement*: To observe how prototypes evolve during training and how they impact classification, we have investigated the changes in the distances between modality prototypes and between class prototypes as the number of training rounds varies respectively. In particular, we calculate the average of the results (i.e., the distance between modality prototypes and the distance between class prototypes) of each batch as the result for one epoch. We use the Euclidean distance as the calculation indicator, and the corresponding results are shown in Fig. 6. From the figure, the distances between modality prototypes are getting smaller, while the distances between class prototypes are getting larger. This indicates that we have successfully maintained the consistency between modalities and increased the separability.

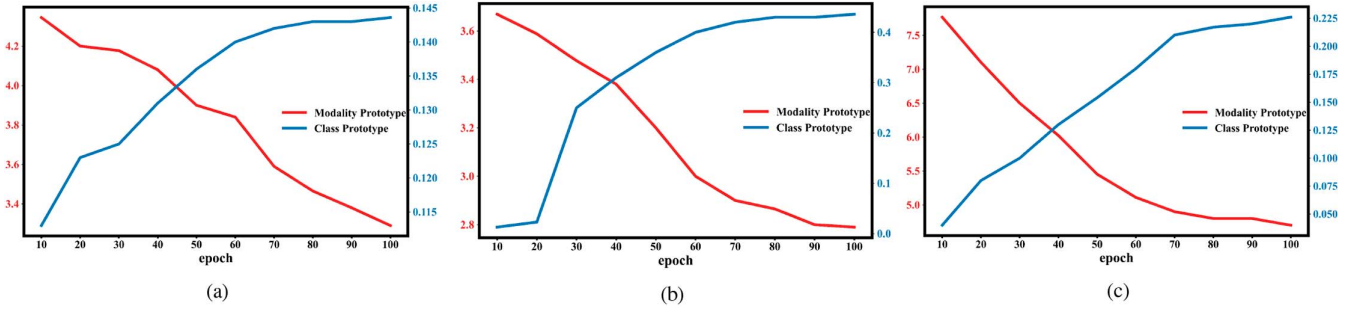


Fig. 6. Variation trend of distances between modality prototypes and between class prototypes on: (a) Weibo; (b) Twitter; and (c) Fakeddit.

TABLE VI
RESULTS ON THE FAKESV DATASET

Method	Accuracy	Fake News			Real News		
		Precision	Recall	F1-score	Precision	Recall	F1-Score
QMFND	0.750	0.769	0.838	0.802	0.714	0.616	0.662
PFBL	0.759	0.791	0.816	0.803	0.706	0.671	0.688

Note: The bold entries denote the best results among multiple compared methods.

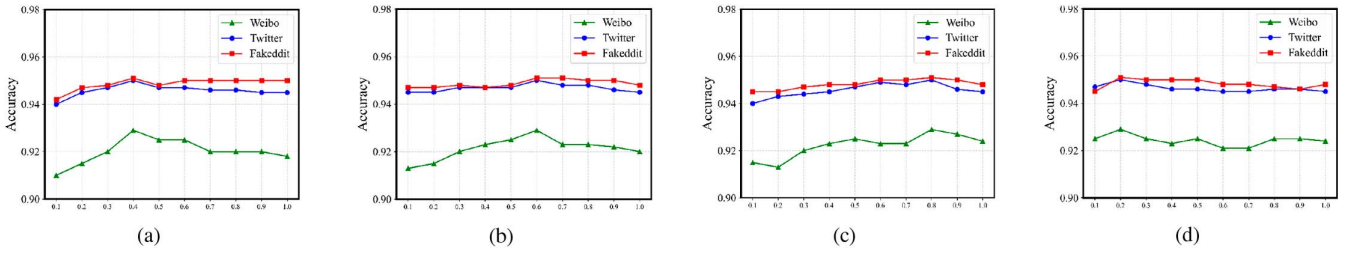


Fig. 7. Performances of PFBL with different values of: (a) λ_1 ; (b) λ_2 ; (c) ε ; and (d) δ on Weibo, Twitter, and Fakeddit.

TABLE VII
COMPUTATIONAL EFFICIENCY AND MEMORY CONSUMPTION OF PFBL ACROSS DIFFERENT DEVICES ON THREE DATASETS

Dataset	Training Time(s)			Testing Time(s)			Memory Consumption (MiB)		
	Weibo	Twitter	Fakeddit	Weibo	Twitter	Fakeddit	Weibo	Twitter	Fakeddit
PFBL (GTX 1080Ti)	44.50	40.14	60.39	27.03	23.64	36.34	2987	3033	3248
PFBL (A6000)	35.20	32.45	51.33	22.98	19.38	30.89	2340	2528	2598

4) *Effects of Model Across More Modalities*: To validate the performance of our model on datasets with more modalities, we choose the FakeSV dataset [33], which stands as the largest publicly accessible Chinese dataset designed for fake news detection on short video platforms. It is characterized by incorporating samples sourced from Douyin and Kuaishou, which are two renowned Chinese short video platforms. It contains 1827 fake news videos, 1827 real news videos. We conduct experiments using three modalities: title, video clips, and keyframes in the dataset, and adopt the standard experimental setting to perform experimental compare. The experimental results are shown in Table VI. From the table, our PFBL can still outperform QMFND in the multimodal fake news detection task containing the video modality.

5) *Parameter Analysis*: To analyze the convergence rates and stability under varying hyperparameters, we conduct experiments to observe hyperparameters λ_1 , λ_2 , ε , δ with different

values on Weibo, Twitter, and Fakeddit. Fig. 7 shows the experimental results of evaluation of λ_1 , λ_2 , ε , δ in the range of [0,1]. From the figure, for the four hyperparameters with different values, the accuracy of the model remains within a stable range on the three datasets, which verifies the stability of our model. The best values of these hyperparameters are $\lambda_1 = 0.4$, $\lambda_2 = 0.6$, $\varepsilon = 0.8$, and $\delta = 0.2$.

6) *Effects of Model Across Different Devices*: To investigate computational efficiency and memory consumption across different devices, we have run our model on two devices with GTX 1080Ti and A6000 GPU. We calculate the training time (for each epoch), testing time (on the total test set) and memory consumption (GPU) on Weibo, Twitter, and Fakeddit. The detailed computational efficiency and memory consumption across different devices of our approach are reported in Table VII. From the table, even though on the device with GTX 1080Ti, the computational efficiency and memory consumption are still acceptable.

V. CONCLUSION

In this article, we propose PFBL, a fully balanced learning approach based on prototype learning to address the dual imbalanced MFND problem. To deal with the modality imbalance problem, we design PMB, which jointly enhances the intraclass compactness of the poorly performing modality as well as restraining the dominant modality in optimization process, such that the optimization processes of dominant modality and the under-performing modality are effectively balanced. To deal with the class imbalance problem, we design PCB, which reduces the intraclass scatter and increases the interclass separability, such that class imbalance is effectively alleviated. Furthermore, we also design a joint prototype updating strategy where we fully consider the interaction between modality prototype and class prototype, which is beneficial to solving the interwined problem.

Experimental results on three datasets, Weibo, Twitter, and Fakeddit demonstrate that PFBL significantly outperforms the state-of-the-art methods. The ablation studies and further analysis validate the effectiveness of the proposed important parts.

Currently, our PFBL is designed for the fully supervised learning case. In real-world social media monitoring systems, a large amount of news may be unlabeled. In the future, PFBL will be extended to the semisupervised version to explore useful information from this unlabeled data.

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